

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283741

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

GOTO; Moe et al.

APPARATUS, METHOD, AND NON-TRANSITORY COMPUTER READABLE MEDIUM

Abstract

Provided is an estimation apparatus comprising an acquisition unit that acquires state data including intensity data of at least one frequency band related to a detection signal detected by a vortex flow meter for a fluid, a state determination estimation unit that estimates a state of the fluid based on the state data acquired by the acquisition unit, a fluid quality estimation unit that estimates, from the intensity data, at least either of a liquid amount or dryness related to the fluid according to the state of the fluid estimated by the state determination estimation unit, and an output unit that outputs at least either of the liquid amount or the dryness estimated by the fluid quality estimation unit.

Inventors: GOTO; Moe (Tokyo, JP), NOMURA; Ryouzuke (Tokyo, JP), OKAMOTO; Kazutoshi (Tokyo, JP), ISHINABE; Atsushi (Tokyo, JP), HAYASHI; Yuichi (Tokyo, JP)

Applicant: Yokogawa Electric Corporation (Tokyo, JP)

Family ID: 94536268

Appl. No.: 18/984772

Filed: December 17, 2024

Foreign Application Priority Data

JP	2024-033633	Mar. 06, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: G01F1/325 (20220101); G01F1/32 (20220101)

U.S. Cl.:

Background/Summary

BACKGROUND

1. Technical Field

[0001] The present invention relates to an apparatus, a method, and a non-transitory computer readable medium.

2. Related Art

[0002] Patent document 1 describes “a measurement method of wetness for measuring a flow rate of the wet vapor distributed in a pipe with one of a vortex flow meter or an ultrasonic flow meter and an orifice flow meter to acquire wetness in the pipe by using the flow rate measurement value”. Patent document 2 describes that “the vortex flow measurement apparatus is capable of measuring, by using a Karman vortex frequency recorded by a vortex flow sensor **200**, a flow speed of a first phase of a medium with two or more phases that usually flows during use, and at the same time is capable of quickly (that is, online) detecting generation of at least a second phase in a form of distributed particle flow and/or a droplet flow”. Patent document 3 describes that “during use of the vortex flow measurement apparatus, a mass flow rate of a wall flow of the second phase is quantitatively measured with a simple method based on a measurement signal as well as a standard deviation or kurtosis decided from the vortex frequency and also referring to the correlation”.

RELATED ART DOCUMENTS

Patent Documents

[0003] Patent Document 1: Japanese Patent Application Publication No. 2013-185915 [0004]

Patent Document 2: Japanese Patent No. 5443514 [0005] Patent Document 3: Japanese Patent No. 5355724

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is an example of a block diagram illustrating a system **10** according to the present embodiment.

[0007] FIG. **2** is an explanatory diagram for explaining a flow regime of a fluid in a pipe **40**.

[0008] FIG. **3** is flow diagram illustrating an example of an operation of a vortex flow meter **100** in a measurement apparatus **20** according to the present embodiment.

[0009] FIG. **4** is an explanatory diagram for explaining intensity data.

[0010] FIG. **5** is a flow diagram illustrating an example of an operation of an estimation apparatus **400** in the measurement apparatus **20** according to the present embodiment.

[0011] FIG. **6** is an explanatory diagram for explaining an example of a first model.

[0012] FIG. **7** is an explanatory diagram for explaining an example of a second model.

[0013] FIG. **8** illustrates an example of a computer **2200** in which a plurality of aspects of the present invention may be embodied wholly or in part.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0014] Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the invention according to claims. In addition, not all of the combinations of features described in the embodiments are essential to the solution of the invention.

[0015] FIG. 1 is an example of a block diagram illustrating a system **10** according to the present embodiment. Note that, these blocks are functional blocks that are each functionally divided, and may not be necessarily required to be matched with actual apparatus configurations. That is, in the present drawing, an apparatus indicated by one block may not be necessarily required to be configured by one apparatus. In addition, in the present drawing, apparatuses indicated by separate blocks may not be necessarily required to be configured by separate apparatuses. The same applies to subsequent block diagrams.

[0016] The system **10** collects data from an instrument installed in a facility such as a plant. The plant includes industrial plants such as chemical or biological plants, as well as plants for managing and controlling well sites or surroundings thereof of gas fields, oil fields or the like, plants for managing and controlling power generation through hydroelectric power generation, thermal power generation, nuclear power generation or the like, plants for managing and controlling energy harvesting through solar power generation, wind power generation or the like, plants for managing and controlling water supply and sewerage, dams or the like. The system **10** comprises a measurement apparatus **20** and an external apparatus **30**.

[0017] The measurement apparatus **20** measures a state of a fluid (a fluid to be measured) that flows inside a pipe **40**. The measurement apparatus **20** comprises a vortex flow meter **100**, a sensor **200**, a computing apparatus **300**, and an estimation apparatus **400**.

[0018] The vortex flow meter **100** detects a vortex that is generated in the fluid to measure a volumetric flow rate or the like of the fluid that flows in the pipe **40**. The vortex flow meter **100** has a vortex generator **110**, a detection unit **120**, a charge converter **130**, an AD converter **140**, a frequency analysis unit **150**, a pulse conversion unit **160**, and a communication unit **170**.

[0019] The vortex generator **110** is provided in the pipe **40** through which the fluid to be measured flows. The detection unit **120** detects at least one detection signal according to the vortex (a Karman vortex) generated by the vortex generator **110** with which the fluid to be measured that flows through the pipe **40** collides. The detection unit **120** may output a first detection signal and a second detection signal each detected by two piezoelectric elements provided at two different places on the vortex generator **110**. Each piezoelectric element outputs, as a detection signal, an electric charge according to stress at the place each piezoelectric element is provided in the vortex generator **110**.

[0020] The charge converter **130** is connected to the detection unit **120**. The charge converter **130** converts the electric charge of at least one detection signal into a voltage signal, thereby converting it into at least one detection signal in a voltage format. The charge converter **130** may convert each of the first detection signal and the second detection signal into a voltage signal. Note that, the charge converter **130** is unnecessary when the detection unit **120** outputs the detection signal in a voltage format or a current format.

[0021] The AD converter **140** is connected to the charge converter **130**. The AD converter **140** converts at least one detection signal in a voltage format into at least one detection signal in a digital format. The AD converter **140** may convert each of the first detection signal and the second detection signal into a digital signal.

[0022] The frequency analysis unit **150** is connected to the AD converter **140**. The frequency analysis unit **150** performs frequency analysis of the detection signal in the digital format input from the AD converter **140**, and performs de-noising of the detection signal based on the frequency analysis. The frequency analysis unit **150** may divide the detection signal into a plurality of frequency bands, and perform frequency analysis based on intensity data representing the signal intensity of each frequency band (as an example, a maximum amplitude or average amplitude of a signal in each frequency band). The frequency analysis unit **150** may decide the frequency band including the vortex frequency of the detection signal and a factor λ for de-noising through the frequency analysis.

[0023] The pulse conversion unit **160** is connected to the frequency analysis unit **150**. The pulse

conversion unit **160** may generate a pulse signal representing a vortex frequency of the detection signal on which de-noising has been performed by the frequency analysis unit **150**.

[0024] The communication unit **170** is connected to the frequency analysis unit **150** and the pulse conversion unit **160**, and is connected to the computing apparatus **300** and the estimation apparatus **400** in a wired or wireless manner. The communication unit **170** is capable of communication with at least one of the computing apparatus **300** or the estimation apparatus **400** using a communication protocol defined by HART (registered trademark), BRAIN, Foundation field bus (registered trademark), ISA **100.11a** or the like. The communication unit **170** may transmit, to the estimation apparatus **400**, state data including the intensity data of at least one frequency band acquired by the frequency analysis unit **150**. The communication unit **170** may transmit, to the computing apparatus **300**, a pulse signal generated by the pulse conversion unit **160**.

[0025] The sensor **200** is connected to the computing apparatus **300** in a wired or wireless manner. The sensor **200** may be, for example, at least one of a pressure sensor, a flow meter, a temperature sensor, a density meter, a viscosity meter, a concentration meter or the like. The sensor **200** may transmit, to the computing apparatus **300**, state data, which is at least one of a flow speed, temperature, pressure, viscosity, concentration, or density measured for the fluid to be measured that flows in the pipe **40**. The sensor **200** may be arranged inside or outside the detection unit **120** of the vortex flow meter **100**, and may transmit the state data to the computing apparatus **300** via the communication unit **170**.

[0026] The computing apparatus **300** is connected to the estimation apparatus **400**. The computing apparatus **300** may be a computer such as a PC (personal computer), a tablet computer, a smartphone, a workstation, a server computer, or a general purpose computer, or may be a computer system in which a plurality of computers are connected. Such a computer system is also a computer in a broad sense. In addition, the computing apparatus **300** may be implemented by one or more virtual computer environments that can be executed in the computer. In addition, the computing apparatus **300** may be an internal computer of the vortex flow meter **100**.

[0027] The computing apparatus **300** calculates one or more state data from the data received from the sensor **200** or the vortex flow meter **100**. The computing apparatus **300** may calculate the state data of the fluid from the pulse signal received from the vortex flow meter **100**. The computing apparatus **300** has a pulse counting unit **310** and a computing unit **320**.

[0028] The pulse counting unit **310** is connected to the computing unit **320**. The pulse counting unit **310** measures a pulse count per unit of time of the pulse signal (that is, the vortex frequency) by performing pulse count of the pulse signal received from the vortex flow meter **100**.

[0029] The computing unit **320** calculates at least one of the pressure, the flow speed, the volumetric flow rate, the density, or the mass flow rate of the fluid from the vortex frequency measured by the pulse counting unit **310**. The computing apparatus **300** outputs, to the estimation apparatus **400**, the state data including at least one of the pressure, the flow speed, the volumetric flow rate, the density, or the mass flow rate calculated by the computing unit **320**. In addition, the computing apparatus **300** may transmit, to the estimation apparatus **400**, the state data including at least one of the flow speed, the temperature, the pressure, the viscosity, the concentration, or the density received from the sensor **200**.

[0030] The estimation apparatus **400** is connected to the external apparatus **30**. The estimation apparatus **400** may be a computer such as a PC, a tablet computer, a smartphone, a workstation, a server computer, or a general purpose computer, or may be a computer system in which a plurality of computers are connected. Such a computer system is also a computer in a broad sense. Also, the estimation apparatus **400** may be implemented by one or more virtual computer environments which can be executed in a computer. In addition, the estimation apparatus **400** may be an internal computer of the vortex flow meter **100**.

[0031] The estimation apparatus **400** estimates at least either of a liquid amount or dryness of the fluid based on the state data related to the fluid detected by the vortex flow meter **100**. The

estimation apparatus **400** comprises an acquisition unit **410**, a state determination estimation unit **420**, a selection unit **430**, a fluid quality estimation unit **440**, a learning processing unit **450**, and an output unit **460**.

[0032] Herein, the fluid in the present embodiment may include at least one gas (vapor) from which water or the like has evaporated or liquid obtained by condensation of gas. In addition, the estimated liquid amount indicates, for example, the mass occupied by the liquid condensed in the vapor. The dryness to be estimated indicates, for example, a ratio of a mass of gas vapor to a mass of vapor.

[0033] The acquisition unit **410** is connected to the communication unit **170** and the computing apparatus **300**. The acquisition unit **410** acquires, from the communication unit **170**, the state data including intensity data of at least one frequency band related to the detection signal detected by the vortex flow meter **100** for the fluid. The acquisition unit **410** may acquire, from the computing apparatus **300**, the state data including at least one of the flow speed, the temperature, the pressure, the viscosity, the concentration, or the density measured for the fluid. The acquisition unit **410** may acquire the state data of at least one of the pressure, the flow speed, the volumetric flow rate, the density, or the mass flow rate of the fluid calculated based on the pulse signal by the computing apparatus **300**.

[0034] The state determination estimation unit **420** is connected to the acquisition unit **410**. The state determination estimation unit **420** estimates the state of the fluid based on the state data acquired by the acquisition unit **410**. The state determination estimation unit **420** may estimate the flow regime of the fluid in the pipe **40** as the state of the fluid based on the state data acquired by the acquisition unit **410**. Note that the flow regime will be described later with reference to FIG. 2. The state determination estimation unit **420** may estimate the state of the fluid based on the state data acquired by the acquisition unit **410** by using a first model for estimating the state of the fluid from the state data.

[0035] The selection unit **430** is connected to the state determination estimation unit **420**. The selection unit **430** selects a computation corresponding to the estimated state of the fluid from a plurality of computations. The selection unit **430** may select a computation different depending on the flow regime of the fluid.

[0036] The selection unit **430** may select a second model corresponding to the state of the fluid estimated by the state determination estimation unit **420** from a plurality of second models for estimating at least either of a liquid amount or dryness related to the fluid from the intensity data of at least one frequency band related to the detection signal of the fluid according to the state of the fluid.

[0037] The fluid quality estimation unit **440** is connected to the selection unit **430**. The fluid quality estimation unit **440** estimates at least either of the liquid amount or dryness related to the fluid from the intensity data acquired by the acquisition unit **410** according to the state of the fluid estimated by the state determination estimation unit **420**. The fluid quality estimation unit **440** may estimate at least either of the liquid amount or dryness related to the fluid from the intensity data acquired by the acquisition unit **410** through the computation selected by the selection unit **430**. The fluid quality estimation unit **440** may estimate at least either of the liquid amount or dryness related to the fluid from the intensity data acquired by the acquisition unit **410** by using the second model selected by the selection unit **430**.

[0038] The learning processing unit **450** is connected to the state determination estimation unit **420**, the selection unit **430**, and the fluid quality estimation unit **440**. The learning processing unit **450** performs learning processing of a plurality of models used in the estimation apparatus **400**. The learning processing unit **450** may generate a first model for estimating the state of the fluid from the intensity data of at least one frequency band related to the detection signal of the fluid. The learning processing unit **450** may generate a second model for each state of the fluid.

[0039] The output unit **460** is connected to the fluid quality estimation unit **440** and the external

apparatus **30**. The output unit **460** outputs, to the external apparatus **30**, at least either of the liquid amount or dryness estimated by the fluid quality estimation unit **440**.

[0040] The external apparatus **30** may be a display apparatus that displays an estimation result of the estimation apparatus **400**, or a PC that stores the estimation result of the estimation apparatus **400** or the like.

[0041] FIG. **2** is an explanatory diagram for explaining the flow regime of the fluid in the pipe **40**. FIG. **2** illustrates a XZ cross section and a YZ cross section in the pipe **40** for each of the stratified flow, the stratified wavy flow, and the annular mist flow, which are the flow regimes. In FIG. **2**, the fluid flows in a Y axis direction in the pipe **40**.

[0042] For example, the fluid containing evaporated water has three flow states (flow regimes) in a state of wet vapor in which liquid water is mixed with gas vapor. In the stratified flow, the liquid flows gently at the bottom of the pipe **40**. In the stratified wavy flow, the liquid flows while creating waves at the bottom of the pipe **40**. In the annular mist flow, the liquid flows in contact with the entire circumference of the inner wall of the pipe **40**, and the number of droplets mixed with the vapor increases. Flow regimes such as these changes according to gas flow speed and wetness of the fluid. The state of the Karman vortex in the measurement of the vortex flow meter **100** and the detection signal that detects the vortex are different for the three flow regimes, and its influence on the measurement related to the fluid is also different. For such a state of the fluid, the same applies to fluids other than the evaporated water.

[0043] Therefore, in the present embodiment, the liquid amount or dryness of the fluid, which is an indicator of vapor quality, is estimated through a computation according to the state of the fluid.

[0044] FIG. **3** is flow diagram illustrating an example of an operation of a vortex flow meter **100** according to the present embodiment. At step S**300**, in the vortex flow meter **100**, the fluid to be measured flows in the pipe **40**, and when the flow speed becomes a certain flow speed or higher, a vortex called a Karman vortex is generated downstream the vortex generator **110**. The Karman vortex is alternately generated on the left and right (or above or below) about the axis of the vortex generator **110**, and the pressure in the fluid is changed by generation of the vortex. The two piezoelectric elements of the detection unit **120** each output the electric charge according to the pressure at the installed place as the first detection signal and the second detection signal.

[0045] At step S**310**, the charge converter **130** converts the first detection signal and the second detection signal, which are electric charge signals, into voltage signals through a charge amplifier. The charge converter **130** may then amplify signal levels of the first detection signal and the second detection signal in the voltage format through a gain amplifier.

[0046] At step S**320**, the AD converter **140** converts, into a digital signal, each of the first detection signal and the second detection signal in an analog format converted into a voltage format by the charge converter **130**.

[0047] At step S**330**, the frequency analysis unit **150** may divide each of the first detection signal and the second detection signal in a digital format into a plurality of frequency bands, and acquire intensity data representing the signal intensity of each frequency band. The frequency analysis unit **150** may divide the detection signal into a plurality of predetermined frequency band according to the inner diameter of the pipe **40** through which the fluid to be measured flows or the kind of the fluid to be measured. For example, the frequency analysis unit **150** may divide the first detection signal into a plurality of frequency bands (for example, an octave band) obtained by dividing the entire frequency band (for example, 0 to 15 KHz), which is the measurement target of the vortex frequency, and acquire the intensity data representing the signal intensity of each frequency band. The frequency analysis unit **150** may divide the second detection signal into a plurality of frequency bands, similarly to the first detection signal, and acquire the intensity data representing the signal intensity of each frequency band.

[0048] For example, the frequency analysis unit **150** may divide the detection signal into a plurality of frequency bands through a bandpass filter. As an example, the frequency analysis unit **150** has a

plurality of bandpass filters associated in a one-to-one manner with the plurality of frequency bands obtained by dividing the entire frequency band which the vortex flow meter **100** targets, and each bandpass filter may have a configuration in which it passes a signal component in the detection signal that is in the assigned passing frequency band, and reduces or removes signal components that is outside the band. Alternatively, the frequency analysis unit **150** may calculate the signal intensity of the detection signal in each frequency band by performing Fourier conversion of the detection signal to convert it into a signal in a frequency region.

[0049] The frequency analysis unit **150** may output, to the communication unit **170**, intensity data of one frequency band among the plurality of divided frequency bands. In addition, the frequency analysis unit **150** may transmit the intensity data of a plurality of divided frequency bands, for example, the intensity data of all frequency bands to the estimation apparatus **400** via the communication unit **170**. The frequency analysis unit **150** may transmit, along with the intensity data, data representing the frequency band corresponding to said intensity data to the estimation apparatus **400** via the communication unit **170**.

[0050] At step S**340**, the frequency analysis unit **150** removes noises in the detection signal by using the intensity data of a plurality of frequency bands to acquire a vortex signal (a signal that generates a frequency that is proportionate to the flow speed generated by the Karman vortex).

[0051] First, the frequency analysis unit **150** identifies a frequency band (also referred to as a vortex frequency band) that includes a vortex frequency of the vortex signal. The frequency analysis unit **150** may identify, as the vortex frequency band, the frequency band including the intensity data with the largest signal intensity in the entire frequency band.

[0052] In addition, the frequency analysis unit **150** identifies the frequency band including the intensity data with the largest signal intensity in the entire frequency band. In the identified frequency band including the largest intensity data, the frequency analysis unit **150** may calculate a signal ratio of said largest intensity data and the intensity data of the other detection signal (that is, the signal intensity ratio of the signal intensity of the first detection signal and the signal intensity of the second detection signal identified frequency band). The frequency analysis unit **150** may identify said frequency band as the vortex frequency band if the calculated signal ratio is within a predetermined range. When the calculated signal ratio is not within the predetermined range, the frequency analysis unit **150** may similarly decide whether the calculated signal ratio is within the predetermined range for the frequency band including the next (in this case, second) largest intensity data. Note that, the predetermined range may be set according to at least one of the frequency band that is presumed to be the Karman vortex generating frequency calculated from the flow speed of the fluid to be measured, the kind of the fluid to be measured, or the inner diameter of the pipe **40** through which the fluid to be measured flows.

[0053] The frequency analysis unit **150** then decides the factor λ for removing noises of the detection signal by using the intensity data of the frequency band other than the identified vortex frequency band. The frequency analysis unit **150** may decide the factor λ for removing the noises by using the intensity data of the frequency band other than the identified vortex frequency band and at least one frequency band that is adjacent before and after said vortex frequency band.

[0054] In the frequency band other than the vortex frequency band, the frequency analysis unit **150** may decide, as the factor λ , a signal ratio that is equal to or more than a predetermined minimum signal ratio among the signal ratios (the ratio of the signal intensity of the first detection signal and the signal intensity of the second detection signal) in the frequency band including the signal intensity that is equal to or more than a predetermined threshold. In this case, the frequency analysis unit **150** may decide the factor λ by using the signal ratio of the frequency band in which a signal intensity of the detection signal, among the first detection signal and the second detection signal, which is the same as the detection signal with the largest signal intensity in the entire frequency band is equal to or more than the predetermined threshold. When there is no signal that satisfies the condition described above, the frequency analysis unit **150** may omit the de-noising,

and output, to the pulse conversion unit **160**, the first detection signal of the vortex frequency band. Note that, the predetermined threshold may be set according to at least one of the kind of the fluid to be measured or the inner diameter of the pipe **40** through which the fluid to be measured flows. In addition, the predetermined minimum signal ratio may be set according to the inner diameter of the pipe **40** through which the fluid to be measured flows.

[0055] The frequency analysis unit **150** then removes noises in the detection signal by using the factor λ decided. The frequency analysis unit **150** may remove the noise through linear combination of the first detection signal A and the second detection signal B output from the AD converter **140** by the factor A. In this manner, the frequency analysis unit **150** can generate a combination signal C (a vortex flow rate signal) including a component of the vortex signal ($C=A-\lambda B$).

[0056] The frequency analysis unit **150** may then extract, from the combined signal, the signal component of the vortex frequency band identified at step S**340** to generate a signal having the vortex frequency. For the combined signal, the frequency analysis unit **150** passes the signal component within the vortex frequency band and reduces or removes signal components that are out of said vortex frequency band by using a bandpass filter, for example. The frequency analysis unit **150** outputs the extracted combined signal of the vortex frequency band to the pulse conversion unit **160**. As an example, this extracted combined signal of the vortex frequency band is a signal that has a shape of an approximately sine wave.

[0057] The frequency analysis unit **150** may transmit the intensity data representing the signal intensity of at least one frequency band for the combined signal as the state data to the estimation apparatus **400** via the communication unit **170**. For example, the frequency analysis unit **150** may divide the combined signal into a plurality of frequency bands that are the same as the plurality of frequency bands divided at step S**330** by using a plurality of bandpass filters with different passing frequency bands. The frequency analysis unit **150** may output, to the communication unit **170** as the state data, at least one of the plurality of pieces of intensity data respectively representing the signal intensity of a plurality of frequency bands obtained by dividing the combined signal. The frequency analysis unit **150** may transmit, along with the intensity data of the combined signal, data representing the frequency band corresponding to said intensity data to the estimation apparatus **400** via the communication unit **170**.

[0058] At step S**350**, the pulse conversion unit **160** may generate a pulse signal having the same frequency as that of the combined signal received from the frequency analysis unit **150**.

[0059] At step S**360**, the communication unit **170** outputs, to the computing apparatus **300**, a pulse signal representing the vortex frequency of the detection signal generated based on the intensity data of at least one frequency band. The communication unit **170** may transmit the pulse signal that is converted into a pulse by the pulse conversion unit **160** to the computing apparatus **300**. In this manner, the computing apparatus **300** can perform pulse count of the received pulse signals, and calculate the frequency according to the number of Karman vortices generated per unit of time.

[0060] FIG. **4** illustrates the signal intensity of the detection signal of the vortex flow meter **100** in a plurality of frequency bands. FIG. **4** illustrates the intensity data representing the signal intensity in each frequency band obtained by dividing each of the first detection signal and the second detection signal in a digital format into a plurality of frequency bands by the frequency analysis unit **150** at step S**330** illustrated in FIG. **3**. In FIG. **4**, the vertical axis indicates the signal intensity, and the horizontal axis indicates the frequency. In FIGS. **4**, A**0** to A**8** each represent the signal intensity of the first detection signal at frequency bands **0-f0**, **f0-f1**, **f1-f2**, **f2-f3**, **f3-f4**, **f4-f5**, **f5-f6**, **f6-f7**, and **f7-f8**, and B**0** to B**8** each represent the signal intensity of the second detection signal at frequency bands **0-f0**, **f0-f1**, **f1-f2**, **f2-f3**, **f3-f4**, **f4-f5**, **f5-f6**, **f6-f7**, and **f7-f8**. The values of the frequencies **f0** to **f8** are each set according to at least one of the inner diameter of the pipe **40** through which the fluid to be measured flows or the kind of the fluid to be measured.

[0061] The frequency analysis unit **150** determines whether the frequency band **f3-f4** including the intensity data A**4** with the largest signal intensity in the entire frequency band **f0-f8** is a vortex

frequency band. The frequency analysis unit **150** calculates the signal ratio ($A4/B4$), and when the calculated signal ratio is within a predetermined range, identifies said frequency band $f3-f4$ as the vortex frequency band.

[0062] The frequency analysis unit **150** decides the factor A for the de-noising by using the intensity data of the frequency band excluding the frequency band $f3-f4$ and the frequency bands $f2-f3$, $f4-f5$, which are before and after the frequency band $f3-f4$. The frequency analysis unit **150** may decide a signal ratio $A8/B8$ to be the factor λ when the signal ratio $A8/B8$ in the frequency band $f7-f8$, among the frequency bands other than the frequency bands $f3-f4$, $f2-f3$, and $f4-f5$, including the signal intensity $A8$, which is equal to or larger than a predetermined threshold, is equal to or larger than a predetermined minimum signal ratio.

[0063] The frequency analysis unit **150** generates a combination signal C through linear combination of the first detection signal A and the second detection signal B output from the AD converter **140** by the factor λ by using the decided factor λ ($C=A-\Delta B$). The frequency analysis unit **150** then divides, by using a plurality of bandpass filters with different passing frequency bands, the combination signal C into a plurality of frequency bands $0-f0$, $f0-f1$, $f1-f2$, $f2-f3$, $f3-f4$, $f4-f5$, $f5-f6$, $f6-f7$, $f7-f8$, and acquires the combination signal $C0$ of the frequency band $0-f0$, the combination signal $C1$ of the frequency band $f0-f1$, the combination signal $C2$ of the frequency band $f1-f2$, the combination signal $C3$ of the frequency band $f2-f3$, the combination signal $C4$ of the frequency band $f3-f4$, the combination signal $C5$ of the frequency band $f4-f5$, the combination signal $C6$ of the frequency band $f5-f6$, the combination signal $C7$ of the frequency band $f6-f7$, and the combination signal $C8$ of the frequency band $f7-f8$.

[0064] The frequency analysis unit **150** outputs the combination signal $C4$ to the pulse conversion unit **160**. The pulse conversion unit **160** generates a pulse signal having a same frequency as that of the combination signal $C4$, and transmits the pulse signal to the computing apparatus **300** via the communication unit **170**. The computing apparatus **300** calculates, from the received pulse signal, the frequency according to the number of Karman vortexes generated per unit of time.

[0065] The frequency analysis unit **150** transmits, to the estimation apparatus **400** via the communication unit **170**, at least one of the intensity data representing the signal intensity of the combination signals $C0-C8$ in a plurality of frequency bands (in particular, the intensity data representing the signal intensity of the combination signal $C4$). The frequency analysis unit **150** may transmit data representing the corresponding frequency band along with the intensity data to the estimation apparatus **400** via the communication unit **170**.

[0066] FIG. 5 is a flow diagram illustrating an example of an operation of the estimation apparatus **400** according to the present embodiment. The estimation apparatus **400** may start an estimation operation upon receiving the state data from the vortex flow meter **100** and the computing apparatus **300**.

[0067] At step S500, the acquisition unit **410** acquires the state data for the fluid to be measured. The acquisition unit **410** acquires, as the state data, the digital signal representing the intensity data of at least one frequency band transmitted from the communication unit **170** of the vortex flow meter **100**. The acquisition unit may acquire the state data including the intensity data of a plurality of frequency bands related to the detection signal of the fluid detected by the vortex flow meter **100**. The acquisition unit **410** may further acquire data representing the corresponding frequency band along with the intensity data. The acquisition unit **410** may acquire the state data including at least one of the flow speed, the temperature, the pressure, the viscosity, the concentration, the density, or the vortex frequency from the computing apparatus **300**. The estimation apparatus **400** uses the state data from the computing apparatus **300** along with the intensity data, enabling improvement in the accuracy of state estimation related to the fluid and the accuracy of estimation of the liquid amount or dryness.

[0068] The acquisition unit **410** may periodically receive the state data. The acquisition unit **410** may output, to the state determination estimation unit **420** and the fluid quality estimation unit **440**,

the plurality of pieces of state data acquired at the same time or at the same cycle from the vortex flow meter **100** and the computing apparatus **300** in association with one another. In the following steps, the state determination estimation unit **420** and the fluid quality estimation unit **440** may perform the estimation for each set of associated state data acquired at the same time or at the same cycle by the acquisition unit **410** from the vortex flow meter **100** and the computing apparatus **300**. [0069] The acquisition unit **410** may perform pretreatment on the acquired intensity data, and output the pretreated intensity data to the state determination estimation unit **420** and the fluid quality estimation unit **440**. The acquisition unit **410** may calculate a combined value of the plurality of pieces of intensity data when intensity data of a plurality of frequency bands is acquired. For example, the acquisition unit **410** may calculate a sum value or average value of intensity data of all or a part of the plurality of pieces of intensity data, and output said sum value or average value of the intensity data to the state determination estimation unit **420** and the fluid quality estimation unit **440**. The acquisition unit **410** may acquire, as the pretreated intensity data, a sum value, a difference, a ratio, or a multiplied value of the maximum intensity data and the intensity data of a frequency band that is adjacent to the frequency band corresponding to said maximum intensity data, among the plurality of pieces of intensity data.

[0070] In addition, the acquisition unit **410** may acquire the intensity data normalized through the pretreatment. The acquisition unit **410** may normalize the intensity data based on reference intensity data. For example, the acquisition unit **410** may calculate a difference or a ratio of the acquired intensity data and the reference intensity data, and acquire the calculated value as the normalized intensity data. Note that, the acquisition unit **410** may have prestored therein the reference intensity data corresponding to each of the plurality of frequency bands, or may use the reference intensity data corresponding to the frequency band of the intensity data to be pretreated. The acquisition unit **410** may use, as the reference intensity data, prestored intensity data at a normal state (for example, the state upon start of the operation) of the vortex flow meter **100**, or the intensity data acquired at a time point before the acquisition of the intensity data to be pretreated. The estimation apparatus **400** can prevent the estimation accuracy from being reduced due to a change in the detection signal caused by individual differences or replacement of parts of the vortex flow meter **100** or the like, by normalizing the intensity data.

[0071] The acquisition unit **410** may output at least one the pretreated intensity data (the pretreated intensity data may also be referred to simply as the intensity data) or the intensity data that is not pretreated to the state determination estimation unit **420**.

[0072] At step S505, the state determination estimation unit **420** determines whether the state of the fluid is a superheated vapor based on the state data of the fluid acquired by the acquisition unit **410**. The state determination estimation unit **420** may determine whether the state of the fluid is a superheated vapor by comparing at least one of the state data acquired by the acquisition unit **410** with a threshold. For example, the state determination estimation unit **420** may determine the state of the fluid to be a superheated vapor (that is, not a wet vapor) in response to the measurement value of the temperature of the fluid exceeding a boiling point of the fluid (a boiling point of the liquid included in the fluid) based on a measurement value of the pressure of the fluid acquired by the acquisition unit **410**. As an example, the state determination estimation unit **420** may acquire, from a vapor curve of the liquid in the fluid, a boiling point corresponding to the measurement value of the pressure of the fluid, and when the measurement value of the temperature of the fluid exceeds the boiling point, determine the fluid to be a superheated vapor. In such a case where the measurement value of the temperature of the fluid exceeds the boiling point of the fluid (Yes at step S505), the estimation apparatus **400** proceeds to step S530, and when the measurement value of the temperature of the fluid is equal to or lower than the boiling point of the fluid (No at step S505), the estimation apparatus **400** may proceed to step S510.

[0073] In addition, the state determination estimation unit **420** may decide that the state of the fluid is a superheated vapor when the signal intensity representing the intensity data is within a

predetermined range. Being within the predetermined range may be to be within a range of signal intensity in a case of a superheated vapor obtained in advance through experiments or the like.

[0074] At step S510, the state determination estimation unit **420** determines whether the state of the fluid is a wet vapor based on the state data of the fluid acquired by the acquisition unit **410**. The state determination estimation unit **420** may determine whether the fluid is a wet vapor based on at least one of the acquired measurement values of the state data by using a wet vapor determination model for estimating whether the fluid is a wet vapor from at least one of the measurement values of the state data. The wet vapor determination model may be a model that outputs a determination result, such as logistic regression, neural network, support vector machine, classification tree, change point detection, k-nearest neighbor, or k-means technique.

[0075] In addition, the state determination estimation unit **420** may determine that the state of the fluid is a wet vapor (that is, not a superheated vapor) in response to the measurement value of the temperature of the fluid included in the state data being less than the boiling point of the fluid. In this case, the state determination estimation unit **420** may determine the state of the fluid by using a vapor curve.

[0076] The state determination estimation unit **420** determines, through step S505 and step S510, whether the state of the fluid is a wet vapor by using the state data acquired by the acquisition unit **410**. The estimation apparatus **400** may proceed to step S520 when the fluid is determined not to be a superheated vapor at step S505 and the fluid is determined to be a wet vapor at step S510 (Yes at S510), and may proceed to step S530 when the fluid is determined not to be a wet vapor (No at S510).

[0077] At step S520, the state determination estimation unit **420** estimates which of a stratified flow, a stratified wavy flow, or an annular mist flow the state of the fluid is based on the state data acquired by the acquisition unit **410**. The state determination estimation unit **420** may estimate the flow regime of the fluid by using a first model for estimating the state of the fluid from the state data related to the fluid. The first model may be a classification model, or may be a model such as logistic regression, neural network, support vector machine, classification tree, change point detection, k-nearest neighbor, or k-means technique.

[0078] The state determination estimation unit **420** may estimate the state of the fluid according to which of the range corresponding to the stratified flow, a range corresponding to the stratified wavy flow, or a range corresponding to the annular mist flow the acquired intensity data belongs to. The range corresponding to the stratified flow, the range corresponding to the stratified wavy flow, and the range corresponding to the annular mist flow may be ranges of signal intensity acquired in advance through experiments for each state.

[0079] At step S525, the selection unit **430** selects a computation according to the state of the fluid estimated by the state determination estimation unit **420** from a plurality of computations for estimating at least either of a liquid amount or dryness. The selection unit **430** may select a second model corresponding to the flow regime of the fluid estimated by the state determination estimation unit **420** from a second model corresponding to the stratified flow, a second model corresponding to the stratified wavy flow, or a second model corresponding to the annular mist flow. The plurality of second models may be generated by learning processing for each corresponding flow regime, and may be different from each other. The selection unit **430** may use a plurality of second models previously generated by the learning processing unit **450**.

[0080] The second model may be a regression model, or may be a model such as multiple regression analysis, neural network, support vector regression, Gaussian process regression, regression tree, autoregressive model, logistic regression, classification tree, change point detection, k-nearest neighbor, or k-means technique. The neural network may include convolutional neural network, recurrent neural network, or long short-term memory neural network. Note that, the second model may be generated through learning processing by using the intensity data representing the signal intensity of a piezoelectric element measured by the vortex flow meter **100**

or the pretreated intensity data as training data, and the dryness or liquid amount corresponding to said intensity data as teacher data.

[0081] The second model may be a regression model that outputs the dryness or liquid amount according to an input of the intensity data of at least one frequency band. As an example, the second model can output the dryness or liquid amount of the fluid to be measured with the intensity data of the frequency band including the vortex frequency as the input. When the fluid (vapor) becomes more wet, since the signal intensity of the detection signal of the piezoelectric element is lowered due to moisture included in the vapor, the signal intensity of the piezoelectric element detected by the vortex flow meter **100** is correlated to the dryness or liquid amount of the fluid to be measured (for example, a linear relationship). Therefore, the second model may be a model that outputs greater dryness or smaller liquid amount according to the input of the intensity data of greater signal intensity.

[0082] At step **S530**, the fluid quality estimation unit **440** estimates at least either of the liquid amount or dryness related to the fluid from the intensity data. The fluid quality estimation unit **440** may estimate at least either of the liquid amount or dryness related to the fluid from intensity data of a plurality of frequency bands. The fluid quality estimation unit **440** may estimate at least either of the liquid amount or dryness related to the fluid from a maximum value of the intensity data of the plurality of frequency bands or a combined value of intensity data of the plurality of frequency bands through pretreatment by the acquisition unit **410**. The fluid quality estimation unit **440** may estimate at least either of the liquid amount or dryness related to the fluid from the intensity data normalized through pretreatment by the acquisition unit **410**.

[0083] The fluid quality estimation unit **440** may define, as the estimation result, at least either of the liquid amount or dryness output in response to the intensity data acquired by the acquisition unit **410** being input to the second model selected by the selection unit **430**. The fluid quality estimation unit **440** may define, as the estimation result, at least either of the liquid amount or dryness output in response to the intensity data of a predetermined frequency band (as an example, the frequency band including the vortex frequency) among the plurality of frequency bands being input to the second model. The fluid quality estimation unit **440** may input, along with the intensity data, data representing the corresponding frequency band to the second model.

[0084] According to the estimated state of the fluid, the fluid quality estimation unit **440** may estimate at least either of the liquid amount or dryness related to the fluid from the acquired intensity data, and the state data including at least one of the flow speed, the temperature, the pressure, the viscosity, the concentration, or the density measured for the fluid. The fluid quality estimation unit **440** may input, to the second model, all kinds of the state data acquired by the acquisition unit **410**, or may input, to the second model, the state data representing a predetermine kind of measurement value among the flow speed, the temperature, the pressure, the viscosity, the concentration, and the density acquired by the acquisition unit **410**, along with the intensity data. The fluid quality estimation unit **440** may estimate one of the liquid amount or dryness, and from said estimated one, calculate the other of the liquid amount or dryness by using the flow rate of the vapor of the fluid.

[0085] The fluid quality estimation unit **440** may estimate that the liquid amount related to the fluid is 0 or that the dryness related to the fluid is 100% when it is determined that the state of the fluid is not a wet vapor by the state determination estimation unit **420** at step **S505** or **S510**. Since it is determined by the state determination estimation unit **420** that the fluid is a superheated vapor or that the fluid is not a wet vapor, the fluid quality estimation unit **440** may define liquid amount 0 or dryness 100% as the estimation result. In this case, the fluid quality estimation unit **440** may calculate, from the state data, enthalpy or heat quantity of the superheated vapor as the estimation result related to the fluid.

[0086] At step **S535**, the output unit **460** outputs the estimation result of the fluid quality estimation unit **440** to the external apparatus **30**. The output unit **460** may output display data for displaying

the estimation result on the external apparatus **30**. In addition, the output unit **460** may output data for notifying a user of the estimation result by an alarm or the like of the external apparatus **30**. As an example, the output unit **460** may output data for notifying the user by an alarm of the external apparatus **30** when the liquid amount or dryness, which is the estimation result, exceeds a predetermined threshold or is equal to or less than the predetermined threshold. The predetermined threshold may be set in advance by the user or the like. In addition, the output unit **460** may output, to the external apparatus **30**, control data for performing the control of a plant according to the estimation result, and the external apparatus **30** may control the plant according to the control data.

[0087] FIG. **6** is an explanatory diagram for explaining an example of the first model. In the example of FIG. **6**, the first model is a model of a neural network, and each node is represented by a circle. In the first model, state data (the vortex frequency, the flow speed, the intensity data, the pressure, and the temperature) related to the fluid acquired from the vortex flow meter **100** and the computing apparatus **300** by the acquisition unit **410** is input to each node of the input layer, and the estimation result of the flow regime of the fluid is output to nodes of the output layer. Three nodes of the output layer include a node to which a computation result related to a stratified flow (for example, the probability that the fluid is a stratified flow) is output, a node to which a computation result related to a stratified wavy flow (for example, the probability that the fluid is a stratified wavy flow), is output, and a node to which a computation result related to an annular mist flow (for example, the probability that the fluid is an annular mist flow) is output.

[0088] Any number of layers and number of neurons may be set for an intermediate layer of the first model. A general nonlinear function such as a ramp function, a sigmoid function, or a hyperbolic tangent function may be set as the activation function of the first model. From the viewpoint of an error backpropagation method, the output layer may be provided with a softmax function at the time of multi-value classification.

[0089] In the first model, in response to the state data being input to the input layer, computation results according to the weight assigned among nodes in the intermediate layer are output to nodes of the output layer, and the state determination estimation unit **420** may output the flow regime with the highest probability as the state of the fluid. Note that, the state determination estimation unit **420** may normalize each value of the state data to a predetermined range and input the normalized value to each node of the input layer. The learning processing unit **450** may perform learning processing by using a set of data of a plurality of measurement values of the state data as the training data and the flow regime related to the measurement value as teacher data and optimize the weight to thereby generate the first model.

[0090] The state data in each flow regime used for the learning processing may be data obtained in advance through experiments, simulations, or operation or the like of a plant. As an example, the state data may be state data in each flow regime acquired via a user or automatically from an experiment apparatus by controlling the flow regime in the pipe **40** through an experiment. In addition, the state data may be acquired for each flow regime decided by the user observing the state of the fluid in an experiment or operation of the plant. In addition, the state data may be acquired for each flow regime identified by the user from a simulation result of the fluid in the pipe **40**, and may be acquired for each flow regime identified automatically from the state of a system being simulated.

[0091] Note that, the first model has at least one of the nodes of the input layer, and all or a part of the plurality of measurement values of the state data acquired by the acquisition unit **410** may be input to the nodes of the input layer. In addition, the wet vapor determination model may be a model similar to the first model, and may be subjected to learning processing similar to that of the first model.

[0092] FIG. **7** is an explanatory diagram for explaining an example of the second model. In the example of FIG. **7**, the second model is a model of a neural network, and each node is represented by a circle. In the second model, state data (the vortex frequency, the flow speed, the intensity data,

the pressure, and the temperature) related to the fluid acquired from the vortex flow meter **100** and the computing apparatus **300** by the acquisition unit **410** is input to each node of the input layer, and the liquid amount or dryness of the fluid is output to nodes of the output layer. Any number of layers and number of neurons may be set for an intermediate layer of the second model. A general nonlinear function such as a ramp function, a sigmoid function, or a hyperbolic tangent function may be set as the activation function of the second model. From the viewpoint of an error backpropagation method, the output layer may be provided with a softmax function at the time of multi-value classification.

[0093] In the second model, in response to the state data being input to the input layer, computation results (the liquid amount or dryness) according to the weight assigned among nodes of the intermediate layer are output to nodes of the output layer, and the fluid quality estimation unit **440** may output said output liquid amount or dryness as the estimation result. Note that, the fluid quality estimation unit **440** may normalize each value of the state data to a predetermined range and input the normalized value to each node of the input layer. The learning processing unit **450** may perform learning processing by using a set of data of a plurality of kinds of state data obtained for each flow regime as training data and optimize the weight to thereby generate the second model. The learning processing unit **450** may generate the second model corresponding to the stratified flow by using, as the training data, a plurality of kinds of state data in a case where the fluid is a stratified flow. The learning processing unit **450** may generate the second model corresponding to the stratified wavy flow by using, as the training data, a plurality of kinds of state data in a case where the fluid is a stratified wavy flow. The learning processing unit **450** may generate the second model corresponding to the annular mist flow by using, as the training data, a plurality of kinds of state data in a case where the fluid is an annular mist flow.

[0094] The state data in each flow regime used for the learning processing may be data obtained in advance through experiments, simulations, or operation or the like of a plant. The state data may be that acquired through a method similar to the state data used for learning processing of the first model.

[0095] Note that, the second model has at least one of the nodes of the input layer, and all or a part of the plurality of kinds of state data acquired by the acquisition unit **410** may be input to the nodes of the input layer.

[0096] The estimation apparatus **400** in the present embodiment has high estimation accuracy since it can estimate the liquid amount or dryness by using a model corresponding to the state of the fluid. In addition, since the estimation apparatus **400** uses the intensity data in a digital format generated by the vortex flow meter **100** for the estimation operation, no additional configuration is required in the estimation apparatus **400** for processing of an analog signal, which enables a simple configuration to process digital values in the estimation operation.

[0097] Note that, the fluid quality estimation unit **440** may estimate the liquid amount or dryness by using a function or a table that defines the relationship between the intensity data and the liquid amount or dryness. In addition, the acquisition unit **410** may not perform pretreatment of the intensity data, in which case, the vortex flow meter **100** may perform pretreatment of the intensity data similar to step S500 and transmit said pretreated intensity data to the estimation apparatus **400**. In this manner, the estimation apparatus **400** can use the pretreated intensity data for the estimation operation.

[0098] In addition, the estimation apparatus **400** may not have the learning processing unit **450**, and the estimation apparatus **400** may have a first model generated by an external learning apparatus and a second model generated by an external learning apparatus prestored therein. In this case, the state determination estimation unit **420** may estimate the state of the fluid by using the stored first model, and the fluid quality estimation unit **440** may estimate at least either of the liquid amount or dryness by using the stored second model. The learning apparatus is only required to include at least the learning processing unit **450** of the present embodiment. The learning apparatus may be

included in a computer such as a PC or an external apparatus such as a cloud.

[0099] In addition, various embodiments of the present invention may be described with reference to flowcharts and block diagrams, wherein the block may serve as (1) a stage in a process in which an operation is performed, or (2) a section of an apparatus having a role of performing an operation. Certain stages and sections may be implemented by dedicated circuit, programmable circuit supplied together with computer readable instructions stored on computer readable media, and/or processors supplied together with computer readable instructions stored on computer readable media. Dedicated circuit may include digital and/or analog hardware circuits, and may include integrated circuits (IC) and/or discrete circuits. The programmable circuit may include a reconfigurable hardware circuit including logical AND, logical OR, logical XOR, logical NAND, logical NOR, and other logical operations, a memory element or the like such as a flip-flop, a register, a field programmable gate array (FPGA) and a programmable logic array (PLA), or the like.

[0100] A computer readable medium may include any tangible device that can store instructions to be executed by a suitable device, and as a result, the computer readable medium having instructions stored thereon includes a product including instructions that can be executed in order to create means for executing operations specified in the flowcharts or block diagrams. Examples of the computer readable medium may include an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, and the like. More specific examples of the computer readable medium may include a floppy (registered trademark) disk, a diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or flash memory), an electrically erasable programmable read-only memory (EEPROM), a static random access memory (SRAM), a compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a Blu-ray (registered trademark) disk, a memory stick, an integrated circuit card, or the like.

[0101] The computer readable instruction may include any of: an assembler instruction, an instruction-set-architecture (ISA) instruction; a machine instruction; a machine dependent instruction; a microcode; a firmware instruction; state-setting data; or either a source code or an object code described in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk (registered trademark), JAVA (registered trademark), C++, or the like, and a conventional procedural programming language such as a “C” programming language or a similar programming language.

[0102] Computer readable instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable processing apparatuses, or to programmable circuit, locally or via a local area network (LAN), wide area network (WAN) such as the Internet, or the like, to execute the computer readable instructions to create means for performing operations specified in the flowcharts or block diagrams. Examples of the processor include a computer processor, a processing unit, a microprocessor, a digital signal processor, a controller, a microcontroller, and the like.

[0103] FIG. 8 illustrates an example of a computer **2200** in which a plurality of aspects of the present invention may be embodied wholly or in part. A program installed in the computer **2200** can cause the computer **2200** to function as an operation associated with the apparatuses according to the embodiments of the present invention or as one or more sections of the apparatuses, or can cause the operation or the one or more sections to be executed, and/or can cause the computer **2200** to execute a process according to the embodiments of the present invention or a step of the process. Such programs may be executed by a CPU **2212** to cause the computer **2200** to perform specific operations associated with some or all of the blocks in the flowcharts and block diagrams described in the present specification.

[0104] The computer **2200** according to the present embodiment includes the CPU **2212**, an RAM **2214**, a graphics controller **2216**, and a display device **2218**, which are interconnected by a host

controller **2210**. The computer **2200** also includes input/output units such as a communication interface **2222**, a hard disk drive **2224**, a DVD-ROM drive **2226**, and an IC card drive, which are connected to the host controller **2210** via an input/output controller **2220**. The computer also includes legacy input/output units such as an ROM **2230** and a keyboard **2242**, which are connected to the input/output controller **2220** via an input/output chip **2240**.

[0105] The CPU **2212** operates according to programs stored in the ROM **2230** and the RAM **2214**, thereby controlling each unit. The graphics controller **2216** acquires image data generated by the CPU **2212** in a frame buffer or the like provided in the RAM **2214** or in itself, such that the image data is displayed on the display device **2218**.

[0106] The communication interface **2222** communicates with other electronic devices via a network. The hard disk drive **2224** stores programs and data used by the CPU **2212** in the computer **2200**. The DVD-ROM drive **2226** reads a program or data from a DVD-ROM **2201** and provides the program or data to the hard disk drive **2224** via the RAM **2214**. The IC card drive reads the programs and the data from the IC card, and/or writes the programs and the data to the IC card.

[0107] The ROM **2230** stores therein boot programs and the like executed by the computer **2200** at the time of activation, and/or programs that depend on the hardware of the computer **2200**. The input/output chip **2240** may also connect various input/output units to the input/output controller **2220** via a parallel port, a serial port, a keyboard port, a mouse port, or the like.

[0108] Programs are provided by a computer readable medium such as the DVD-ROM **2201** or the IC card. The programs are read from the computer readable medium, are installed in the hard disk drive **2224**, the RAM **2214**, or the ROM **2230** which is also an example of the computer readable medium, and are executed by the CPU **2212**. The information processing described in these programs is read by the computer **2200**, and provides cooperation between the programs and the various types of hardware resources. The apparatus or method may be configured by implementing operations or processings of information according to use of the computer **2200**.

[0109] For example, in a case where communication is performed between the computer **2200** and an external device, the CPU **2212** may execute a communication program loaded in the RAM **2214** and instruct the communication interface **2222** to perform communication processing based on a processing written in the communication program. Under the control of the CPU **2212**, the communication interface **2222** reads transmission data stored in a transmission buffer processing area provided in a recording medium such as the RAM **2214**, the hard disk drive **2224**, the DVD-ROM **2201**, or the IC card, transmits the read transmission data to the network, or writes reception data received from the network in a reception buffer processing area or the like provided on the recording medium.

[0110] In addition, the CPU **2212** may cause the RAM **2214** to read all or a necessary part of a file or database stored in an external recording medium such as the hard disk drive **2224**, the DVD-ROM drive **2226** (DVD-ROM **2201**), the IC card, or the like, and may execute various types of processing on data on the RAM **2214**. Then, the CPU **2212** writes the processed data back in the external recording medium.

[0111] Various types of information such as various types of programs, data, tables, and databases may be stored in a recording medium and subjected to information processing. The CPU **2212** may execute, on the data read from the RAM **2214**, various types of processing including various types of operations, information processing, conditional judgement, conditional branching, unconditional branching, information retrieval/replacement, or the like described throughout the present disclosure and specified by instruction sequences of the programs, and writes the results back to the RAM **2214**. In addition, the CPU **2212** may retrieve information in a file, a database, or the like in the recording medium. For example, when a plurality of entries, each having an attribute value of a first attribute associated with an attribute value of a second attribute, is stored in the recording medium, the CPU **2212** may retrieve, out of the plurality of entries, an entry with the attribute value of the first attribute specified that meets a condition, read the attribute value of the second attribute

stored in said entry, and thereby acquiring the attribute value of the second attribute associated with the first attribute meeting a predetermined condition.

[0112] The programs or software modules described above may be stored in a computer readable medium on or near the computer **2200**. In addition, a recording medium such as a hard disk or an RAM provided in a server system connected to a dedicated communication network or the Internet can be used as a computer readable medium, thereby providing a program to the computer **2200** via the network.

[0113] While the present invention has been described with the embodiments, the technical scope of the present invention is not limited to the above-described embodiments. It is apparent to persons skilled in the art that various alterations and improvements can be added to the above-described embodiments. It is also apparent from description of the claims that the embodiments to which such alterations or improvements are made can be included in the technical scope of the present invention.

[0114] It should be noted that the operations, procedures, steps, stages, and the like of each process performed by an apparatus, system, program, and method shown in the claims, the specification, or the drawings can be realized in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the output from a previous process is not used in a later process. Even if the operation flow is described using phrases such as “first” or “next” for the sake of convenience in the claims, specification, or drawings, it does not mean that the process must be performed in this order.

Claims

1. An estimation apparatus comprising: an acquisition unit that acquires state data including intensity data of at least one frequency band related to a detection signal detected by a vortex flow meter for a fluid; a state determination estimation unit that estimates a state of the fluid based on the state data acquired by the acquisition unit; a fluid quality estimation unit that estimates, from the intensity data, at least either of a liquid amount or dryness related to the fluid according to the state of the fluid estimated by the state determination estimation unit; and an output unit that outputs at least either of the liquid amount or the dryness estimated by the fluid quality estimation unit.
2. The estimation apparatus according to claim 1, wherein the acquisition unit acquires a digital signal representing the intensity data of at least one frequency band from the vortex flow meter.
3. The estimation apparatus according to claim 1, wherein the acquisition unit acquires the state data including at least one of a flow speed, a temperature, a pressure, a viscosity, concentration, or density measured for the fluid, and the fluid quality estimation unit estimates, from the intensity data acquired by the acquisition unit, and the state data including at least one of the flow speed, the temperature, the pressure, the viscosity, the concentration, or the density measured for the fluid, at least either of the liquid amount or dryness related to the fluid, according to a state of the fluid estimated by the state determination estimation unit.
4. The estimation apparatus according to claim 3, wherein the vortex flow meter outputs, to a computing apparatus, a pulse signal representing a vortex frequency of the detection signal generated based on the intensity data of at least one frequency band, and the acquisition unit acquires the state data including at least one of the pressure, the flow speed, a volumetric flow rate, the density, or a mass flow rate of the fluid calculated based on the pulse signal by the computing apparatus.
5. The estimation apparatus according to claim 1, wherein the acquisition unit acquires the state data including intensity data of a plurality of frequency bands related to a detection signal of the fluid detected by the vortex flow meter, and the fluid quality estimation unit estimates at least either of the liquid amount or dryness related to the fluid from the intensity data of the plurality of

frequency bands.

- 6.** The estimation apparatus according to claim 5, wherein the fluid quality estimation unit estimates, from a maximum value among the intensity data of the plurality of frequency bands or a combined value of intensity data of the plurality of frequency bands, at least either of the liquid amount or dryness related to the fluid.
- 7.** The estimation apparatus according to claim 1, wherein the acquisition unit acquires the intensity data that is normalized, and the fluid quality estimation unit estimates, from the intensity data that is normalized, at least either of the liquid amount or dryness related to the fluid.
- 8.** The estimation apparatus according to claim 1, further comprising: a selection unit that selects, from a plurality of computations, a computation corresponding to the state of the fluid estimated by the state determination estimation unit, wherein the fluid quality estimation unit estimates, from the intensity data acquired by acquisition unit, at least either of the liquid amount or dryness related to the fluid through the computation selected by the selection unit.
- 9.** The estimation apparatus according to claim 1, wherein the state determination estimation unit estimates, based on the state data acquired by acquisition unit, which of a stratified flow, a stratified wavy flow, or an annular mist flow the state of the fluid is.
- 10.** The estimation apparatus according to claim 9, wherein the state determination estimation unit: determines whether the state of the fluid is wet vapor based on the state data acquired by the acquisition unit; and estimates which of a stratified flow, a stratified wavy flow, or an annular mist flow the state of the fluid is based on the state data acquired by the acquisition unit, when the state of the fluid is determined to be wet vapor.
- 11.** The estimation apparatus according to claim 1, wherein the state determination estimation unit estimates, based on the state data acquired by the acquisition unit, the state of the fluid by using a first model for estimating the state of the fluid from the state data.
- 12.** The estimation apparatus according to claim 8, wherein the selection unit selects, from a plurality of second models for estimating, from intensity data of at least one frequency band related to a detection signal of the fluid, at least either of a liquid amount or dryness related to the fluid, a second model corresponding to the state of the fluid estimated by the state determination estimation unit according to the state of the fluid, and the fluid quality estimation unit estimates, from the intensity data, at least either of the liquid amount or dryness related to the fluid by using the second model selected by the selection unit.
- 13.** The estimation apparatus according to claim 12, further comprising: a learning processing unit that generates the second model for each state of the fluid.
- 14.** A learning apparatus comprising; a learning processing unit that generates, from intensity data of at least one frequency band related to a detection signal detected by a vortex flow meter for a fluid, a second model for estimating at least either of a liquid amount or dryness related to the fluid for each state of the fluid.
- 15.** A method comprising: acquiring, by a computer, state data including intensity data of at least one frequency band related to a detection signal detected by a vortex flow meter for a fluid; estimating, by the computer, a state of the fluid based on the state data acquired; estimating, by the computer from the intensity data, at least either of a liquid amount or dryness related to the fluid according to the state of the fluid estimated; and outputting, by the computer, at least either of the liquid amount or the dryness estimated.
- 16.** A learning method comprising: generating, by a computer from intensity data of at least one frequency band related to a detection signal detected by a vortex flow meter for a fluid, a second model for estimating at least either of a liquid amount or dryness related to the fluid for each state of the fluid.
- 17.** A non-transitory computer readable medium having recorded thereon a program that causes a computer to function as: an acquisition unit that acquires state data including intensity data of at least one frequency band related to a detection signal detected by a vortex flow meter for a fluid; a

state determination estimation unit that estimates a state of the fluid based on the state data acquired by the acquisition unit; a fluid quality estimation unit that estimates, from the intensity data, at least either of a liquid amount or dryness related to the fluid according to the state of the fluid estimated by the state determination estimation unit; and an output unit that outputs at least either of the liquid amount or the dryness estimated by the fluid quality estimation unit.

18. A non-transitory computer readable medium having recorded thereon a learning program that causes a computer to function as: a learning processing unit that generates, from intensity data of at least one frequency band related to a detection signal detected by a vortex flow meter for a fluid, a second model for estimating at least either of a liquid amount or dryness related to the fluid for each state of the fluid.
